Lecture 3 ERS312 Notes

How did captains determine longitude

- Based on Greenwich time as a reference
- Clocks at the time were pendulum driven which was a problem
- Clock is set at Greenwich time which is the reference time (0 meridian)
- 360/24
- In an hr the earth rotates counterclockwise 15 degrees every hour
- Greenwich is ahead of us by 5 hours (so how many degrees in longitude) * 15x5= 75 degrees
- 9:52 am in Greenwich
- Noon to 9.52am = 2 hours and 8 minutes (we are at noon)
- Must be East of Greenwich
- 2 hours= 30 degrees
- 8 mins= 2 degrees *8 min= 0.133 hours so 15x0.133
- So 32 degrees E

- Southern hemisphere winter is June (complete darkness 24hours)
- 24 hour darkness in June or December
- Tropic of Capricorn and characteristics: 23.5 degrees S; mirror image of tropic of cancer in southern hemisphere (sun in zenith position *sun is directly above) *march and sep 22

Dead reckoning

- A way to determine a position based on a previous position
- Have an hourglass you turn (30min)
- Need to know speed and time course
- Can determine next position based on what you know of previous position
- Speed was measured at back of the ship by tossing a line with a log at the end (drags in the water) *knots were counted to know how much rope was passed out *speed in ocean measured in knots because of this

Problems:

- Does not take in effect of current and wind
- If wind/currents push ship off course

Ocean Exploration

- First evidence from archeologists
- Cave has evidence of human activity in the ocean
- Collected marine snails (snail has a door *operculum)
- Going in ocean due to food scarcity

Global Temp Anomalies

- Had ups and downs of warm and cold temperatures
- Time of ice ages
- People had little food resources

Pacific Navigators

- Started 5000 BC
- Ventured to pacific from new guinea